Project Proposal: Retail Sales Dashboard

Data Analytics / Business Intelligence Project
Instructor: Jessie Melendres

1. Project Identity

Project Title: RetailPulse: A Sales Dashboard for Identifying Top Products and Revenue-Driving Categories

Chosen Focus Areas: Problem Framing, Data Pipeline, SQL Basics

Repository Link: ____________________________

2. Problem Statement

Retail businesses generate large volumes of transactional data every day, but without a structured way to analyze it, decision-makers often rely on gut feeling rather than evidence when deciding which products to restock, promote, or discontinue. Sales data is frequently scattered across spreadsheets or point-of-sale exports, making it difficult to quickly answer basic but critical questions such as: Which products generate the most revenue? Which categories are underperforming? Are there seasonal or time-based sales patterns being missed?

Manually aggregating this data in spreadsheets is slow, error-prone, and doesn't scale as transaction volume grows. Existing off-the-shelf dashboard tools are often expensive or overly generic, not tailored to a specific retail business's product hierarchy or reporting needs. This project addresses that gap by building a purpose-built sales dashboard, backed by a proper SQL data pipeline, that surfaces top-performing products and categories at a glance.

3. Target Users

Primary users: Retail business owners and store/branch managers who need to make fast, data-informed decisions on inventory, promotions, and product mix.

Secondary users: Sales and marketing staff who want visibility into which categories are driving performance, to better plan campaigns and stock priorities.

This tool shifts users from manually compiling sales reports in spreadsheets to opening a live dashboard that automatically surfaces top products, top categories, and revenue trends — reducing reporting time from hours to seconds and enabling faster, more confident business decisions.

4. Data Source(s)

Data Type: Structured transactional data (CSV/SQL tables) — sales records, product catalog, and category mapping.

Source: A public retail sales dataset (e.g., a Kaggle retail/e-commerce transactions dataset) covering product-level sales across multiple categories and dates, used to simulate a real retail business's transaction history.

Volume: Estimated at least 1,000–5,000+ transaction records spanning multiple product categories, sufficient to demonstrate meaningful trends, top-seller rankings, and category-level aggregation.

5. Success Criteria

1. **Data Accuracy:** The data pipeline correctly loads, cleans, and aggregates 100% of source records with no duplication or data loss.

2. **Query Performance:** Core SQL queries (top products, top categories, revenue trends) return results in under 2 seconds on the full dataset.

3. **Insight Utility:** The dashboard correctly identifies and visually ranks the top 10 products and top 5 categories by revenue.

4. **Clarity:** At least 3 test users can identify the top revenue-driving product/category within 10 seconds of viewing the dashboard, without additional explanation.

5. **Reproducibility:** The SQL data pipeline can be re-run on a refreshed dataset and produce updated results without manual rework.

6. Team Composition

As per course requirements, each teammate must own a clear pipeline component.

Team Member Name	Primary Responsibility	Assigned Component
Angel Frederick Paredes	Lead Developer	Problem Definition & KPI Design, Data Cleaning, Transformation & Loading, SQL Query Design & Database Schema, Dashboard Design, Testing & Documentation
Jeeruz Marl Tabar	Research Support	Compiled candidate datasets and reference sources; reviewed data definitions
Luke Kibry Jao	Testing Support	Manual testing of dashboard queries and outputs across sample scenarios
Clein Lord Will Surigao	Documentation Review	Reviewed checkpoint write-ups and the final report